

YIGITER TUNCER

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SUMMARY

I'm finishing my master's in Mechatronics & Robotics at Politecnico di Milano, after a mechanical engineering degree at Koç University. I've worked and interned across a range of fields so far — aerospace manufacturing, automotive R&D (Mercedes-Benz), cloud data-center operations (AWS) and more as stated below — and adapted quickly to each one, which I think comes down to my genuine curiosity about subjects complementary to my core engineering background. Lately that curiosity of mine has also expanded to Python and machine learning, which I'm picking up through coursework and my thesis (reconstructing 3D CAD models from 2D drawings), on top of my existing CAD/FEA, robotics, and manufacturing background. I take my work seriously, get along easily with people, and would like to work on a hands-on, dynamic team.

WORK EXPERIENCE

DCEO Internship, Amazon Web Services (AWS)

August 2025 - October 2025

- Assisted in maintaining critical mechanical and electrical facility infrastructure, ensuring operational performance standards in areas like safety and availability for AWS data centers.
- Contributed to troubleshooting facility and rack-level events and supported root cause analysis of equipment failures, maintaining high uptime for mission-critical servers.
- Performed data extraction and conducted sensor gap analysis to ensure the integrity and completeness of critical facility monitoring data.

R&D Homologation Engineer, Mercedes-Benz Turk A.S

July 2023 - September 2024

- Contributed to the certification process for vehicle compliance, ensuring all technical and legal standards were met for road-worthiness and safety.
- Gained expertise in vehicle mechanisms under Daimler AG, supporting the development and testing phases of truck manufacturing.
- Assisted in interpreting and applying global regulatory standards, aiding system and whole vehicle type approval processes to ensure product compliance across TR, EU, and Brazil markets.

Summer Intern, TUSAS (Turkish Aerospace)

June 2022 - July 2022

- Participated in the assembly of the TF-X jet aircraft, gaining hands-on experience in advanced manufacturing processes.

Summer Intern, TurkTraktor

August 2022 - September 2022

- Developed paint-shop KPI reports and supported 5S/Kaizen efficiency initiatives; engaged in robotic operations within the DÜRR paint shop.

Summer Intern, ARM Aluminium

June 2021 - August 2021

- Supported press-extrusion machine operation and production planning, including material dispatch and intake.

EDUCATION

Master of Science: Mechanical Engineering, Mechatronics & Robotics

Politecnico di Milano, Milano

**September 2024 -
December 2026 (expected)**

Bachelor of Science in Mechanical Engineering

Koc University, Istanbul

September 2019 - March 2024

Saint Benoît French Highschool

Istanbul

2014 - 2019

PROJECTS

3D Reconstruction of Parametric CAD Models from 2D Engineering Drawings (MSc Thesis, Politecnico di Milano) — February 2026 – Nov 2026 (Ongoing)

- Designed a modular hybrid deep learning pipeline (YOLOv11, U-Net/Transformer-based vectorization, GraphSAGE, prefix-tuned Transformer decoder) with a cross-view graph attention fusion module and OCR cross-validation middleware to reconstruct editable, GD&T-compliant parametric 3D CAD models from 2D engineering drawings, integrated into FreeCAD via its Python API.

Automated Helmet Defect Inspection & Marking

Feb 2026 – Jun 2026

- Designed and fully scripted (Python, TM12 Listen Node) a robotic workcell that uses 3D stereo vision (ZED Mini camera) to detect surface defects on a motorcycle helmet and physically mark them, including cyclic visor actuation — with no offline teaching, with my project group.

Reinforcement Learning & ML-Based Control for Logistics Automation and Compliant Robotic Systems — February 2026 - Jun 2026

- Trained a PPO reinforcement learning agent in MATLAB for logistics parcel singulation, and designed (with partial Python implementation) an ML-driven control framework to extend a compliant robot's reachable workspace by exciting its passive degree of freedom.

**Topology Optimization Of The Hub Carrier Of An Ultra-Efficient Lightweight Battery Electric Vehicle
September 2025**

- Utilized FEA-based topology optimization techniques to redesign a vehicle hub carrier, significantly reducing mass while adhering to critical stiffness and structural integrity constraints.

EXTRACURRICULAR

Member of SPARK Team, Ku Air (AI And Robotic) Society

August 2021- August 2023

- Contributed to the mechanical and design teams for the TEKNOFEST Robotaxi Full Autonomous Vehicle Competition.

Undergraduate R.A, KU Mechanical Characterization Lab

March 2021 - September 2023

- Researched nano-electromechanical systems under Dr. Erdem Alaca, working alongside PhD candidate Sina Zare Pakzad.

ADDITIONAL INFORMATION

- Technical Skills:** Siemens NX, SolidWorks, CATIA, Ansys Workbench, AutoCAD, MATLAB, MATLAB Simulink, LabView, COMSOL, Python, MS Office, Mathematica, Canva, Arduino IDE, Overleaf, Altair HyperWorks, Github, PyTorch, FreeCad
- Languages:** English (Fluent), French (B1), Turkish (Native)
- Accomplishments:** Technofest21, Robotaxi-Full Scale Autonomous Vehicle Competition Finalist (finished as 6th out of 30 team), Barcelona U16 International Basketball Tournament MVP – July 2016, U18 Basketball Serie B League Istanbul Champion- March 2018, "Wheels in Motion: Empowering Mobility with Motorized Wheelchairs", Graduation Project